Area of Interest (1): Blended Wing Body (BWB) and C-17 Propulsion

The Air Force Operational Energy Office (SAF/IE), is seeking concepts for a dual-use propulsion system for both military and commercial aircraft. The North American Classification System (NAICS) Code and associated description is 336412 - Aircraft Engine and Engine Parts Manufacturing.

Current and potential future aircraft in both sectors require engines with 35,000-50,000-lbs of thrust; however, no current production capability exists for propulsion systems in this thrust class. This opportunity represents a potential military/commercial market of $35-40 billion dollars through 2045.

The most recent National Defense Strategy directed the military services to prioritize energy demand reduction to reduce logistics requirements in contested environments. Subsequent National Defense Authorization Acts (NDAA) have highlighted the same and authorized significant funding to achieve those objectives. While for different reasons, alignment between the military and commercial sectors in this instance provides an opportunity to leverage common objectives for accelerated advancement of a new propulsion system or consequential upgrades and program life-extension to existing propulsion systems in the 35,000-50,000-lbs thrust category. Both traditional engine original equipment manufacturers (OEMs) and non-traditional companies are encouraged to submit concepts.

Military applications: Military applications for a 35,000-50,000-lbs thrust engine under evaluation may include a re-engine, or technology upgrade, for the US Air Force's 221 C-17 Globemaster transport aircraft, as well as any potential new military platforms based upon blended wing body (BWB) technology.

Despite a fleet age spanning 11-32 years, utilization of the highly versatile C-17 Globemaster continues to exceed expectations. As the C-17 will remain a mainstay of the Air Force's strategic airlift capabilities for at least the next three decades, a re-engine or technology upgrade for the current F117 (PW2040) engines could provide value.

With contested logistics risk reduction as a key priority in recent National Defense Authorization Acts, the Air Force is pursuing innovative technologies for its future capabilities. The Department of the Air Force (DAF) recently partnered with industry to develop a prototype aircraft to demonstrate the capabilities of BWB technology. With a design that differs from a traditional tube-and-wing aircraft, the BWB blends the aircraft body into its high-aspect-ratio wing, decreasing aerodynamic drag by at least 30% and providing additional lift. This increased efficiency will enable extended range, more loiter time, and increased payload delivery efficiencies, capabilities that are vital to mitigating fuel logistics risks. Several military configurations are possible with the BWB, including cargo, aerial refueling and bomber or arsenal aircraft. Together, these aircraft types account for approximately 60% of the Air Force's total annual jet fuel consumption. As outlined in the fiscal year 2023 NDAA the Department of Defense (DoD) plans to invest $235 million from fiscal year 2023 through 2026 to fast-track the development of this transformational dual-use technology, with additional private investment expected. The effort is the result of collaboration between the Department of the Air Force (DAF) and the National Aeronautics and Space Administration (NASA), with assistance from the DoD's Office of Strategic Capital.

Commercial applications: With significant investments required to maintain and modernize its fleet across airlift, fighter, and drone platforms, the Air Force seeks to leverage synergistic overlap with the civil aviation industry wherever possible. Civil aviation demand is increasingly trending towards a mid-size, mid-range aircraft. Effectively addressing this civil aviation mid-market opportunity will require an engine in the 35,000-50,000-lbs thrust class, coincident with the Air Force's consideration of future C-17 propulsion options and potential BWB applications. Prospective engine candidates could include present commercial solution.

With respect specifically to the Air Force investment in BWB technology: the commercial industry, including passenger airlines and air freight companies, stand to benefit from the development of BWB technology as well, most critically in decreasing operational fuel costs and meeting increasingly stringent international emissions standards. The BWB configuration presents the first step-change improvement in aerodynamic efficiency in multiple decades and is one of the most promising prospects for commercial aviation to achieve significant increases in efficiency. In consultation with the commercial aviation community, including a number of prominent prospective airline customers, a middle-market sized aircraft has been identified as an attractive entry-point for the BWB family to commercial aviation. A middle-market BWB platform requires a 35,000-50,000-lbs thrust engine. Consensus is that demand for a BWB aircraft, delivering 40%-50% fuel burn improvement versus current generation aircraft, could significantly exceed [20] aircraft per month by the middle of the next decade.

The Air Force and commercial aviation community seek to leverage the intersection of their requirements to accelerate the development of new aircraft platforms and their corresponding propulsion systems.

High-Level Concept Requirements:

a. **Turbofan Engine:** A turbofan engine rated at 35,000-50,000-lbs thrust at Sea-level static, capable of delivering the necessary power and efficiency for a wide range of aircraft applications.
b. **Dual-Use Capable:** The engine shall be capable of, and eligible for, dual military and commercial use, allowing for flexibility in deployment and application.
c. **Emissions Compliant**: The engine shall be compliant with all current and known future emissions standards.
d. **Timely Entry into Service:** Entry into service no later than 2030; if this timeline cannot be met, please provide an expected availability date to ensure planning and scheduling can be adjusted accordingly.
e. **Cost-Share:** A cost-share arrangement, or other means of project financing, for the development and initial manufacturing phases, where the developer and manufacturer will share the costs and risks associated with bringing the engine to market, and reap the benefits of a successful partnership. Unique or unconventional methods of project financing are encouraged.

f. **Production Ramp-Up Plan:** A detailed plan for ramping up production to meet the demanding delivery schedule, which includes:
    i. Enabling delivery of no less than 40 engines per month plus spares (20 aircraft per month) by 2035, and
    ii. Delivering no less than 80 engines per month plus spares (40 aircraft per month) by 2037, to ensure a steady and increasing supply of engines to meet growing demand.

g. **Innovative Development Approach:** Open to unique and innovative development approaches, including new materials, designs, and manufacturing techniques, to reduce costs, improve efficiency, and enhance performance.

h. **Life-Cycle Cost Reduction:** Identification of pathways to reduce life-cycle costs, including maintenance, repair, and overhaul (MRO) costs, fuel consumption, and other operating expenses, to ensure the engine provides long-term value and affordability for operators and owners.

This AOI is governed by CSO FA4452-25-S-C001.

SOLUTION BRIEF DUE DATE AND TIME: Solution briefs will be accepted until Thursday 13 March 2025, 1400 CST. Solution briefs may be submitted at any time during this period. Solution briefs received after specified date and time, will be considered late in accordance with 52.212-1(f).

Submission POC: Solution briefs shall be submitted to the Contracting Points of Contact (POCs):

Agreements/Contracting Officer
Name: Lisa Pendragon
Phone: 505-225-9752
Email: lisa.pendragon@us.af.mil

Agreements/Contract Specialist
Name: Madison Tikalsky
Phone: 618-256-9962
Email: madison.tikalsky.1@us.af.mil

STEP ONE: SOLUTION BRIEF

SUBMISSION INFORMATION:

Submission Limit: Solution Briefs shall not exceed five written pages (not including title page or table of contents) using 12-point Times New Roman font, 1-inch margins (on all sides) or, alternatively, Solution Briefs may take the form of briefing slides which shall not exceed 15 slides (not including title slide). Any pages or briefing slides exceeding the limit will be removed and not evaluated. Companies may submit multiple unique Solution Briefs in either form described above.

Required Content:

1. Section A: Company Information

    a. Title of Project
    b. Name of Company
    c. Business Size
    d. Company's Commercial and Government Entity (CAGE) number
    e. Unique Entity Identifier (UEI) number
    f. Contracting Point of Contact (POC) [Name, Phone, Email]
    g. Technical POC [Name, Phone, Email]

2. Section B: Technical Solution

Describe how your proposed propulsion system concept meets the following requirements. We are interested in unique and innovative development approaches, including new materials, designs, and manufacturing techniques, to reduce costs, improve efficiency, and enhance performance.

a. **Turbofan Engine:** A turbofan engine rated at 35,000-50,000-lbs thrust at Sea-level static, capable of delivering the necessary power and efficiency for a wide range of aircraft applications.
b. **Dual-Use Capable:** The engine shall be capable of, and eligible for, dual military and commercial use, allowing for flexibility in deployment and application.
c. **Emissions Compliant**: The engine shall be compliant with all current and known future emissions standards.
d. **SAF-Compatible:** The engine shall be capable of burning traditional Jet A fuel and up to 100% sustainable aviation fuel (SAF) for commercial applications.
e. **Timely Entry into Service:** Entry into service no later than 2030; if this timeline cannot be met, please provide an expected availability date to ensure planning and scheduling can be adjusted accordingly.
f. **Cost-Share:** A cost-share arrangement, or other means of project financing, for the development and initial manufacturing phases, where the developer and manufacturer will share the costs and risks associated with bringing the engine to market, and reap the benefits of a successful partnership. Unique or unconventional methods of project financing are encouraged.
g. **Production Ramp-Up Plan:** A detailed plan for ramping up production to meet the demanding delivery schedule, which includes:
    i. Enabling delivery of no less than 20 aircraft per month (40 engines per month plus spares) by 2035, and
    ii. Delivering no less than 40 aircraft per month (80 engines per month plus spares) by 2037, to ensure a steady and increasing supply of engines to meet growing demand.
h. **Life-Cycle Cost Reduction:** Identification of pathways to reduce life-cycle costs, including maintenance, repair, and overhaul (MRO) costs, fuel consumption, and other operating expenses, to ensure the engine provides long-term value and affordability for operators and owners.

3. Section C: Product Roadmap and Development Plan

    a. Product roadmap, highlighting any currently planned products that could meet the requirements described in the introduction.
    b. If nothing on your product roadmap would satisfy these requirements, identify a product that could be modified or developed in this time frame to satisfy the requirements.
        i. High-level description of the current product and the modifications you would propose and/or would be required.
    c. Timeline and rough-order estimate of the cost for such a development effort or upgrade.
    d. Provide evidence that non-federal sources will contribute at least one-third (1/3) of the total prototype project cost.

Offerors are responsible for monitoring SAM.gov for any amendments to this solicitation that may have updated information. All solution briefs shall be valid for 180 calendar days after the submission date.

Solution Brief Evaluation and Selection Process: Solution briefs will be evaluated against the factors as described in CSO FA4452-25-S-C001.
Solution briefs will be categorized as Selectable or Not Selectable as follows:

1. Selectable: Solution briefs are recommended for advancement to Step Two.
2. Not Selectable: The solution brief is not recommended to advance to Step Two.

Those Companies whose Solution Brief are evaluated to be Selectable will enter deliberations for the collaboration on Concept of Design Milestones for the inclusion in the Step Two deliverables. At that time a due date and time for Step Two: Detailed Approach – Concept of Design will be communicated.

STEP TWO: DETAILED APPROACH – CONCEPT OF DESIGN

SUBMISSION INFORMATION:

The Company will be invited to develop and submit a full written Detailed Approach and negotiate appropriate terms and conditions governing the prototype project. Companies may discuss ideas and details of the approach during the writing process with the Government.

1. Section A shall provide the technical approach;
    a. Title Page
        i. Company Name
        ii. Title
        iii. Point of Contact Name (Title, Date, E-Mail Address, Phone, and Address)
        iv. Any subcontractors or additional team members
        v. An abstract which provides a concise description of the approach
    b. Technical Approach
        i. Describe the background and objectives of the proposed solution proposed

in Step 1, the proposed technical approach and the resources needed to execute it.

ii. Include the nature and extent of the anticipated results.

iii. Include any potential ancillary and operational issues such as certifications, algorithms, and any engineering/software development methodologies to be used.

iv. This approach shall include a Statement of Work (SOW) identifying the work to be performed and the deliverables. Provide a detailed project schedule that outlines the various phases of work to be accomplished within the proposed period of performance. You may refer to the Solution Brief that prompted this step, but do not duplicate it.

c. Government Support Required

i. Identify the type of support, if any, the Company requests of the Government in general such as facilities, equipment, data, and information, and/ or materials.

2. Section B shall address the price schedule portions of the approach.
   a. The Company shall propose a total price to complete the project and provide any other data or supporting information that is necessary for the determination of a fair and reasonable price. This can include, but is not limited to, commercial price catalog or other proprietary information to help the Government assess project price.

Companies may propose their own internal terms and conditions (e.g., Service License Agreements (SLA) and/or User License Agreements (ULA)) to be considered during deliberations with the Government. Companies should note that there are certain terms and conditions the Government is unable to accept. However, projects awarded through the CSO are flexible to adopt customary industry standards where it is otherwise legal and where it meets the Government's needs.

Detailed Approach Review and Evaluation Process: Each detailed approach will be evaluated by the Government and shall stand on its own technical merit. The Detailed Approach will be evaluated in accordance with the following evaluation factors:

1. Technical
2. Importance to agency
3. Price

All evaluation factors are of equal importance.

Method of Evaluation: Scientific, technological, and/or other subject-matter experts.

Detailed Approaches will be categorized as Acceptable or Not Acceptable as follows:

1. Acceptable: Detailed Approaches are recommended for acceptance if sufficient funding is available.
2. Not Acceptable: Even if funds were available, the Detailed Approach should not

be funded.